Name ____ Date ___________ Period _____

Absolute Value Inequalities: Special Cases and Interval Notation

Algebra II Honors | Unit 2 | Day 8 | TEK A2.6.F

Objective: I can write absolute value inequality solutions in interval notation, and recognize both special cases: no solution, and all real numbers.

Vocabulary

No solution — an isolated absolute value set < or ≤ a negative number.

All real numbers — an isolated absolute value set > or ≥ a negative number - true no matter what x is.

Part 1 — Interval Notation Reference

Inequality	Compound form	Interval notation
$\lvert x\rvert > 5$	________________________	____________________
$\lvert x\rvert \ge 5$	________________________	____________________
$\lvert x\rvert < 5$	________________	____________
$\lvert x\rvert \le 5$	________________	____________

Part 2 — Variable on Both Sides

Example. $|3x + 1| - 4 \le 2x - 1$

Isolate: $|3x + 1| \le 2x + 3$ → $-(2x+3) \le 3x+1 \le 2x+3$ → _________________

Part 3 — Two Special Cases, Side by Side

A. No solution. $4 + |x + 1| \le 2$

Isolate: $|x + 1| \le$ ________ An absolute value can never be ___________________________________. Solution: ______________________

B. All real numbers. $4 + |x + 1| \ge 2$

Isolate: $|x + 1| \ge$ ________ An absolute value is ALWAYS ____. Solution: ___________________________________

These two came from the SAME left side, just a flipped inequality symbol. In one sentence, explain why one has no solution and the other is always true.

Sample: ____

Practice

1. Solve $|x - 2| + 6 \le 1$ ______________________
2. Solve $|x - 2| + 6 \ge 1$ ___________________________
3. Solve $|2x - 1| > x + 5$ _____________________________

Exit Ticket

Solve both $3 + |x - 4| < 1$ and $3 + |x - 4| > 1$. Explain in one sentence why they come out so differently despite looking almost identical.

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